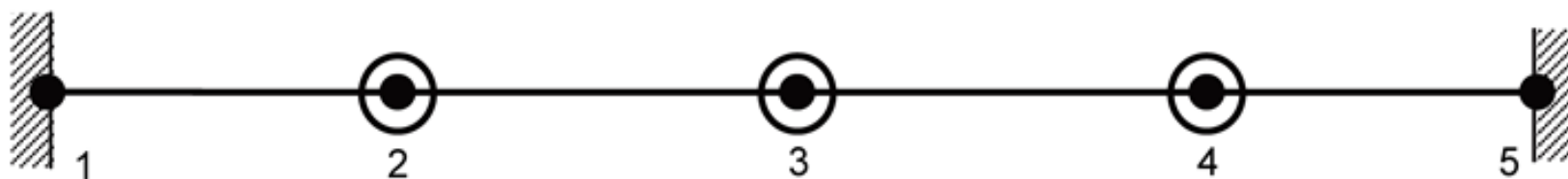
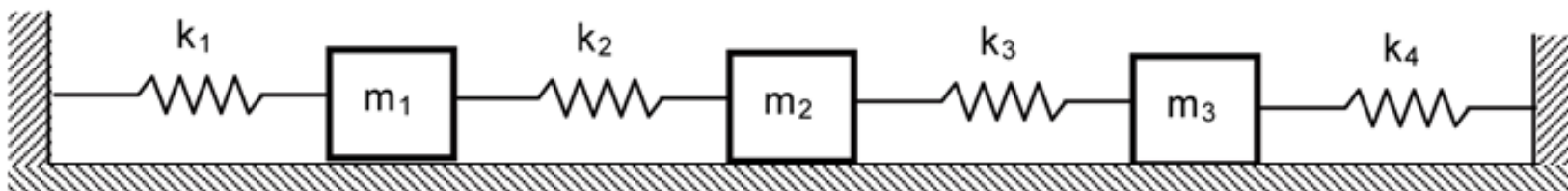
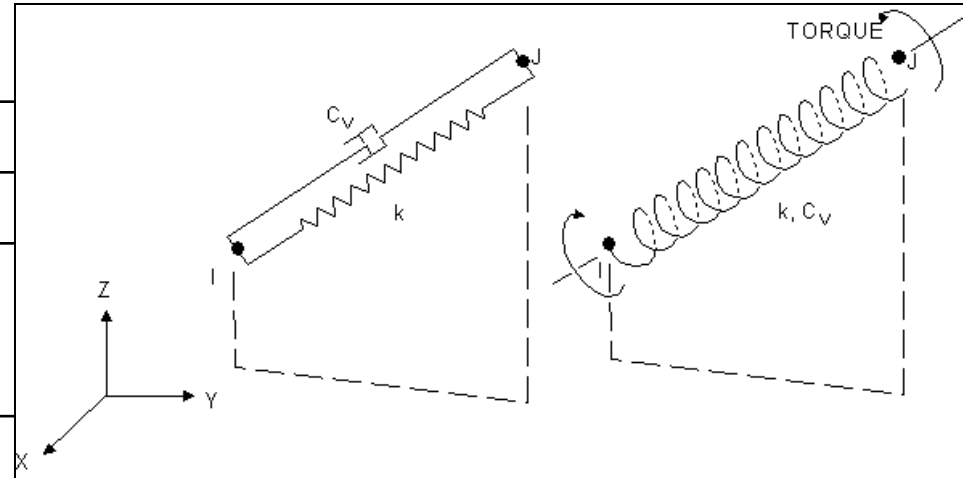


三個質點彈簧系統之振動頻率。 $m_1 = 1 \text{ kg}$ 、 $m_2 = 2 \text{ kg}$ 、 $m_3 = 3 \text{ kg}$ ，質點間隔為 1m ； $k_1 = 1000 \text{ N/m}$ 、 $k_2 = 2000 \text{ N/m}$ 、 $k_3 = 2000 \text{ N/m}$ 、 $k_4 = 1000 \text{ N/m}$ ，用Block Lanczos方法求取前三個自然頻率。



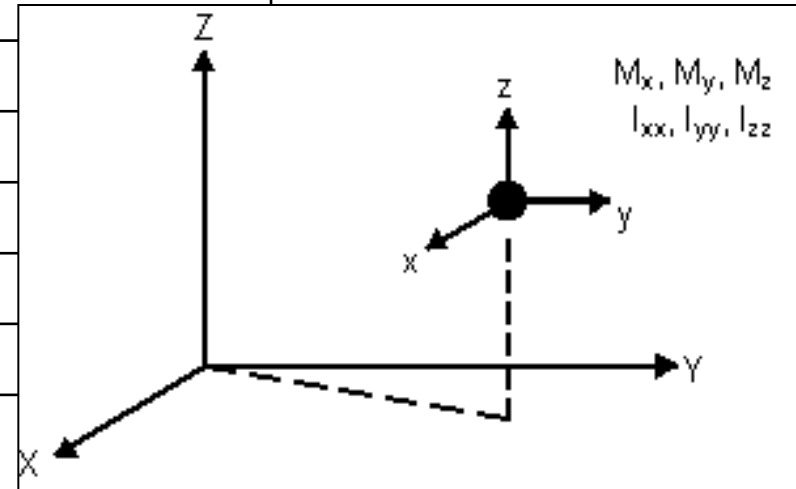
COMBIN14: Spring/Damper

Element Name	COMBIN14
Nodes	I, J
Degrees of Freedom	UX, UY, UZ if KEYOPT(3) = 0 ROTX, ROTY, ROTZ if KEYOPT(3) = 1 UX, UY if KEYOPT(3) = 2 etc.
Real Constants	K, CV1, CV2
Material Properties	None
Surface Loads	None
Body Loads	None
Special Features	Nonlinear (if CV2 is not zero), Stress stiffening, Large deflection, etc.
KEYOPT(1)	0 - Linear Solution (default) 1 - Nonlinear solution (required if CV2 is non-zero)
KEYOPT(3)	0 - 3-D longitudinal spring-damper 1 - 3-D torsional spring-damper 2 - 2-D longitudinal spring-damper (2-D elements must lie in an X-Y plane)



MASS21: Structural Mass

Element Name	MASS21
Nodes	1
Degrees of Freedom	UX, UY, UZ, ROTX, ROTY, ROTZ if KEYOPT(3) = 0 UX, UY, UZ if KEYOPT(3) = 2 UX, UY, ROTZ if KEYOPT(3) = 3 UX, UY if KEYOPT(3) = 4
Real Constants	MASSX, MASSY, MASSZ, IXX, IYY, IZZ if KEYOPT(3) = 0 MASS if KEYOPT(3) = 2 MASS, IZZ if KEYOPT(3) = 3 MASS if KEYOPT(3) = 4
Material Properties	None
Surface Loads	None
Body Loads	None
Special Features	Large deflection
KEYOPT(2)	Key for element coordinate system
KEYOPT(3)	0 - 3-D mass with rotary inertia 2 - 3-D mass without rotary inertia 3 - 2-D mass with rotary inertia 4 - 2-D mass without rotary inertia



○ 指令說明 ◀◀

TYPE, *ITYPE*

指定元素種類編號。

REAL, *NSET*

指定元素特性參數編號。

MAT, *MAT*

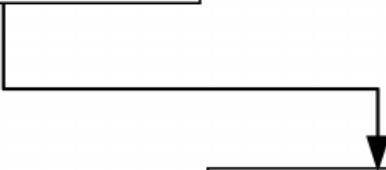
指定材料性質編號。

ITYPE, NSET, MAT 皆是指編號，預設均為 1。

Element Type 1
Element Type 2
Element Type 3

Real Constant 1
Real Constant 2
Real Constant 3
Real Constant 4

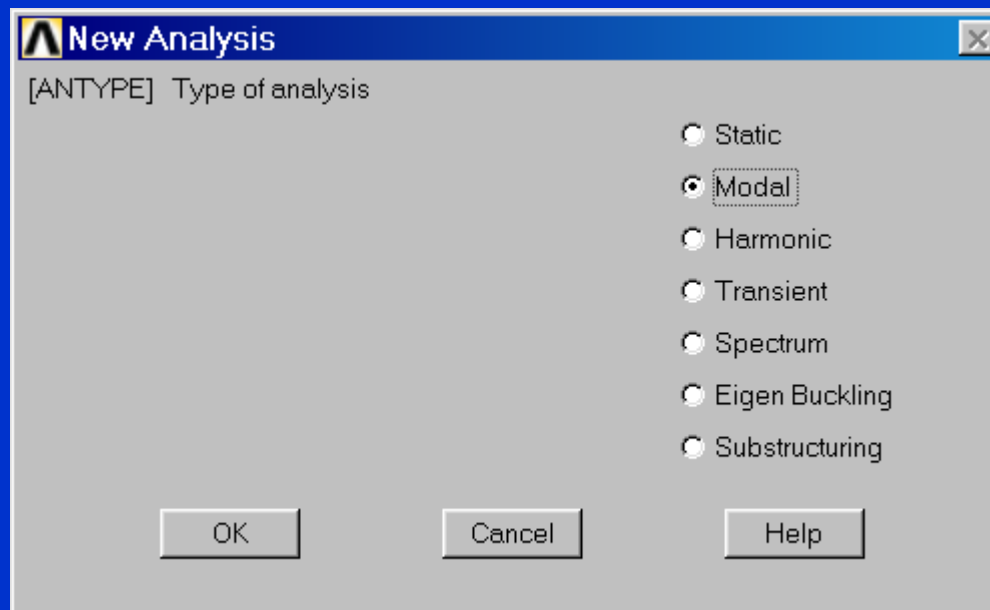
Material Property 1
Material Property 2



Element Attribute	TYPE,1	REAL,1	MAT,1
Element Attribute	TYPE,1	REAL,2	MAT,1
Element Attribute	TYPE,2	REAL,3	MAT,2
Element Attribute	TYPE,3	REAL,4	MAT,2

Analysis Type

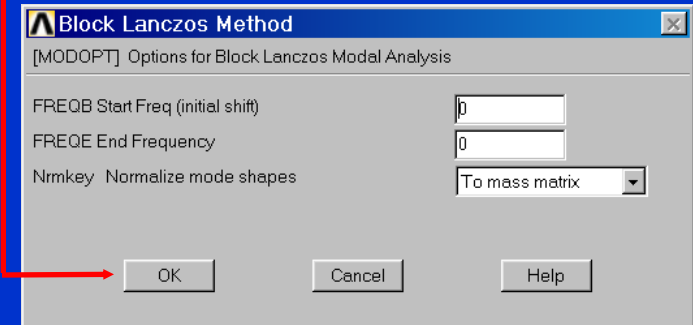
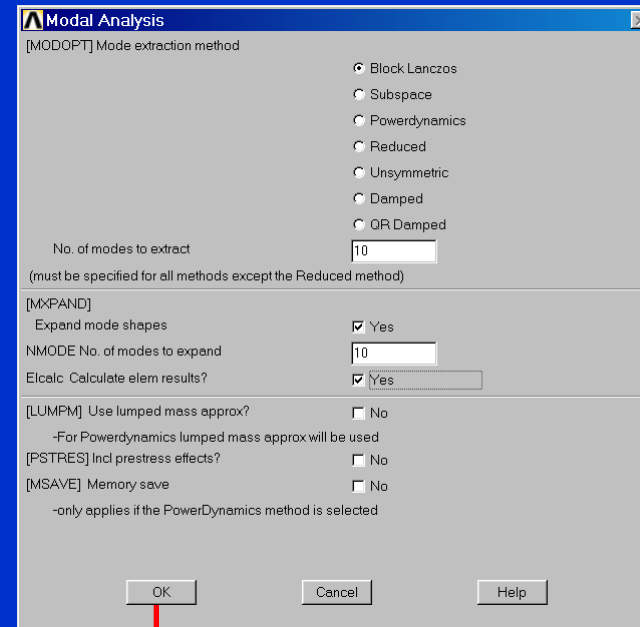
- The analysis type is *modal*.
 - Solution > Analysis Type > New Analysis
 - Or issue `ANTYPE,MODAL`



Modal Analysis ...Procedure

Analysis Options

- **Mode extraction options**
 - **MODOPT** command or **Solution > Analysis Type > Analysis Options**
 - **Method: Block Lanczos is generally recommended.**
 - **Number of modes: Specify the number of modes you want to extract.**
- **Mode expansion options**
 - **MXPAND** command or **Solution > Analysis Type > Analysis Options**
 - **Expanding modes** allows you to view them in the postprocessor.
 - **Number of modes is usually the same as the number extracted.**
- **Prestress effects**
 - **PSTRES** command or **Solution > Analysis Type > Analysis Options**
 - **calculate mode shapes of a prestressed structure, such as a spinning turbine blade.**



Default settings are usually sufficient